

UNIVERSITY OF LIBERAL ARTS BANGLADESH

Department of Media Studies and Journalism

RESEARCH PROPOSAL

Algorithmic News Feeds and Misinformation Susceptibility Among Young Urban News Consumers in Dhaka: A Mixed-Methods Study

Submitted by

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Abstract

Most urban Bangladeshi university students now get most of their news through algorithmically ranked social media feeds. What we don't know, with any quantitative precision, is how that exposure relates to the everyday skill of telling reliable news from misinformation. Most of the empirical work on this question comes from Western contexts; the few Bangladesh-specific studies are qualitative, single-site, or focused on the supply side rather than on individual susceptibility. This study aims to produce the first comparative quantitative dataset on the relationship between algorithmic feed reliance and misinformation discernment among Dhaka university students, alongside a qualitative account of how

students themselves describe their credibility-evaluation practices.

The design is cross-sectional and mixed-methods, and will be pre-registered on the Open Science Framework before main-phase data collection. We recruit 300–360 students across six universities in Dhaka — three private (ULAB, BRAC, North South University) and three public (Dhaka University, Jahangirnagar University, Bangladesh University of Engineering and Technology) — stratified by faculty within each site. The quantitative phase measures news consumption across three modalities (Facebook feed, YouTube feed, and AI-mediated news exposure, operationally defined in section 6.3), three indices of misinformation discernment (the validated MIST-20 as the primary outcome, an adapted shortened AOT scale and a contextually constructed Bangladesh headline task as robustness checks), share-intent ratings on the headline items as a partial behavioural measure, and four socioeconomic controls beyond institution type. The qualitative phase consists of 18 semi-structured interviews (three per site) covering credibility-evaluation practices, awareness of algorithmic curation, and barriers to source verification.

Our primary hypothesis is that heavier reliance on algorithmic feeds is associated with lower misinformation discernment; whether media literacy training and socioeconomic background moderate this association is examined as a pre-specified exploratory question. The design is correlational; it cannot establish causality, and findings should not be generalised beyond Dhaka university populations. What it can do is produce a comparative dataset, a set of locally adapted instruments, and a qualitative account of student reasoning that other researchers and Bangladeshi media-literacy programmes can build on.

1. Introduction

On 21 August 2024, a photograph of a woman tied to a tree circulated widely on Facebook across South Asia. Major Indian outlets reposted it as evidence of communal violence against a Hindu school principal in Dhaka, naming her as Geetanjali Barua, Principal of Azimpur Government Girls' School and College. The claim was false. The woman in the photograph was not the principal. By the time anyone said so in print, the image had reached millions of users through the algorithmic feed.

I reported on that incident for *The Business Standard* (Hossain & Chowdhury, 2024). It was not an isolated case. In my subsequent fact-checking work with FactWatch I have seen the same pattern repeatedly — most recently a viral video falsely captioned as a BJP attack on a Kolkata mosque, which was in fact footage of a party office being vandalised in West Bengal (FactWatch, 2026) — and I have watched university students share such items as fact. What I saw, in real time, was factually incorrect narratives spreading faster than any correction the local fact-checking infrastructure could mount. Those observations are what made me want to do this study. They are not, by themselves, evidence for a generalisable claim.

It is an empirical puzzle: does the pattern I have seen across these news cycles show up systematically among ordinary news consumers, and if so, who is most affected?

The case for asking is at least quantitative. According to DataReportal's *Digital 2026: Bangladesh* report, Bangladesh had 64.0 million social media user identities in October 2025 — 36.3 percent of the total population — with Facebook the dominant platform at approximately 72.5 million accounts (Kemp, 2025). In traditional media environments, editorial gatekeeping filters information before publication, however imperfectly. In platform environments, ranking is done by algorithms designed to maximise engagement. Content that provokes outrage, fear, or moral shock tends to spread further than content that is accurate or corrective. On Twitter at least, Vosoughi et al. (2018) demonstrated that "falsehood diffused significantly farther, faster, deeper, and more broadly than the truth in all categories of information" (p. 1146); in their bot-removal analyses, "robots accelerated the spread of true and false news at the same rate" (p. 1146), implying the differential is driven primarily by human re-sharing. Whether that pattern generalises to Bangladeshi news consumers using Facebook, YouTube, and increasingly AI-mediated news exposure is what this study asks.

There are reasons to think Bangladesh is a particularly important site for the question. Platform penetration is high, media literacy infrastructure is thin, and the dominant ranking systems are designed, deployed, and governed outside the country by companies with limited direct accountability to Bangladeshi users or norms. Sajadieh et al. (2026), in the Stanford AI Index 2026, report that four in five U.S. high school and college students now use generative AI for school-related tasks, suggesting a trajectory that comparable urban Bangladeshi populations are likely to follow even without good local statistics.

What's missing is empirical work. Peer-reviewed quantitative research examining the relationship between algorithmic feed exposure and measurable misinformation susceptibility in Bangladesh is, on a systematic JSTOR, Scopus, and Google Scholar search (terms: misinformation, Bangladesh, algorithm, social media, conducted January 2026), effectively absent. This study is an attempt at that work: a mixed-methods, two-phase, six-site comparison of private and public university students in Dhaka, with quantitative measurement of platform reliance and misinformation discernment followed by interviews that ask students to describe their own credibility-evaluation practices.

The design is cross-sectional and correlational. It cannot establish causality. What it can do is produce the first comparative dataset of its kind for Bangladeshi university students and a set of locally adapted instruments that other studies can borrow.

2. Research Objective

To examine associations between algorithmically curated social media news consumption and misinformation susceptibility among university students in Dhaka, comparing private and public university samples, and to explore the social and cognitive mechanisms that participants themselves identify as shaping their epistemic behaviour in platform-mediated information environments.

3. Research Questions

Primary research questions

RQ1. Is heavier reliance on algorithmically curated social media news consumption associated with lower misinformation discernment, as measured by the MIST-20, among university students in Dhaka?

RQ2. Do specific consumption patterns — disaggregated by modality (Facebook feed, YouTube feed, AI-mediated news exposure), by frequency, and by self-reported awareness of algorithmic curation — differ in their association with discernment?

RQ3. Do discernment patterns differ between private and public university students, and to what extent do socioeconomic and digital-capital variables account for any observed institutional difference?

RQ4. To what extent do scores on the MIST-20, an adapted shortened AOT scale, and a contextually constructed Bangladesh headline task converge as measures of discernment in this sample?

Qualitative sub-questions

RQ5. How do young news consumers describe their own processes for evaluating the credibility of news content encountered through social media feeds?

RQ6. What barriers to critical news evaluation do participants identify, and what role do they assign to platform design, educational background, and social environment in shaping their epistemic habits?

Hypotheses

The quantitative phase tests four confirmatory hypotheses. All tests are two-tailed at $\alpha = .05$ unless stated otherwise; the corresponding models are specified in section 6.3.

H1 (primary; RQ1). The composite algorithmic-feed-reliance index will be negatively associated with MIST-20 overall discernment, controlling for the covariates in Model 1. The confirmatory test is the reliance coefficient in Model 1.

H2 (secondary; RQ2). When the composite index is replaced by modality-specific scores (Model 2), Facebook feed reliance and YouTube feed reliance will each be negatively associated with MIST-20 discernment; these two tests form one Holm-corrected family. AI-mediated exposure is included in Model 2 as a pre-specified exploratory modality, with no directional prediction, because its expected direction is theoretically ambiguous: assistant use may displace verification or may itself be a verification practice.

H3 (secondary; RQ3). Institution type will be associated with discernment before socioeconomic adjustment; the confirmatory question is whether the institution-type coefficient remains distinguishable from zero after the four socioeconomic / digital-capital controls are added (Model 1 vs Model 3; non-directional).

H4 (measurement; RQ4). The MIST-20, the adapted AOT scale, and the Bangladesh headline task will converge at pairwise $r \geq .40$, triggering the pre-specified composite supplementary outcome; pairwise $r < .40$ triggers separate analysis under the convergence-handling rule (section 6.3).

Moderation analyses (prior media literacy training $\times$ reliance; parental education $\times$ reliance), share-intent analyses, algorithmic-awareness contrasts, and the AI-exposure sub-component composite are pre-specified but designated exploratory: at $N = 300$ the design has 80% power only for single-interaction increments of $f^2 \geq$ approximately .027, which exceeds typical attenuated interaction effects. No confirmatory claims will be made from exploratory analyses.

4. Literature Review

Keywords: *Algorithmic news curation · platform misinformation · media literacy · epistemic vulnerability · digital news consumption · Bangladesh*

4.1 The platform turn in news consumption — and what we now know and don't

Social media platforms have displaced traditional media institutions as the primary news gateway for large segments of the global population: the Reuters Institute's 48-market survey finds that engagement with "traditional media sources such as TV, print, and news websites continues to fall, while dependence on social media, video platforms, and online aggregators grows" (Newman et al., 2025, p. 10). In Bangladesh, the Digital 2026: Bangladesh report records 64.0 million social media user identities in October 2025, 36.3 percent of the total population, with Facebook the dominant platform at approximately 72.5 million accounts and growth of around 8.5 million users (15.3 percent) year on year (Kemp, 2025). The same report puts TikTok at 56.2 million adult users, YouTube at 49.8 million, and Instagram at only 9.15 million — 5.2 percent of the population — so Bangladeshi platform news exposure runs overwhelmingly through Facebook and video feeds rather than the Instagram-centred ecosystems prominent in Western samples (Kemp, 2025). No robust Bangladesh-specific, platform-by-platform news-consumption survey yet exists; this gap is part of what the present study addresses.

This shift is not just a change in distribution channel. Editorial decisions about newsworthiness and reliability are made — however imperfectly — by human journalists and editors operating within professional norms. Platform ranking is made by algorithmic systems designed to optimise for user engagement.

Early scholarship described this shift in strong filter-bubble or echo-chamber terms (Pariser, 2011; Sunstein, 2017). The empirical record since has substantially complicated that picture. Bakshy et al.'s (2015) Facebook-internal study concluded that "compared with algorithmic ranking, individuals' choices played a stronger role in limiting exposure to cross-cutting content" (p. 1130). Eady et al. (2019), using linked survey and Twitter data, found "a substantial amount of overlap (51%) in the ideological

distributions of accounts followed by users on opposite ends of the political spectrum" (p. 1) — echo-chamber exposure less severe than the popular framing suggested. Most recently, the 2023 Meta-collaboration studies (Guess et al., 2023; Nyhan et al., 2023) found surprisingly small effects of algorithm changes on political attitudes over weeks-long intervention windows; Nyhan et al. (2023) report that reducing like-minded content "had no measurable effects on eight preregistered attitudinal measures" (p. 137). Bagchi et al. (2024), in a *Science* eLetter critiquing the Guess et al. (2023) study, argued that Meta's emergency algorithm changes around the 2020 U.S. election contaminated the Guess et al. baseline, meaning the "standard" algorithm in that study was already less misinformation-promoting than usual. Metzler and Garcia (2024) review the broader evidence and conclude that the role of algorithms in misinformation and polarisation is "far from straightforward" (p. 735) and that substantial further empirical research is needed.

The takeaway for this study is that the literature is more contested than the popular framing suggests. Algorithmic feeds plausibly contribute to misinformation exposure, but individual-level factors (motivated reasoning, partisanship, prior beliefs, analytical thinking) also matter substantially (Pennycook & Rand, 2019; Bail et al., 2018). The relationship between exposure and behaviour is bidirectional and mediated by individual differences. This study is designed to examine the association without presupposing a single causal direction.

4.2 Algorithmic ranking and misinformation propagation

A substantial body of research documents that false or low-quality content propagates differently on platforms than accurate content does. Vosoughi et al. (2018), in their study of Twitter rumour cascades, found that false news spread faster and further than true news — and that controlled analyses attributed the difference primarily to human re-sharing rather than to bots (in their experimental conditions, bots accelerated true and false content at roughly equal rates). Brady et al. (2017) demonstrated, also on Twitter, that moral-emotional words in political messages "increased their diffusion by a factor of 20% for each additional word" and that this "moral contagion was bounded by group membership" (p. 7313) — systematic, human-driven amplification of emotionally provocative content.

The mechanism on engagement-ranked feeds therefore appears to be a feedback loop: human users react more strongly to certain kinds of content; the algorithm picks up on the engagement signal and surfaces similar content more often. Whether the algorithm's contribution to that loop is "small but consequential" or "the main driver" is contested. Guess et al. (2023) found that restructuring Facebook's feed away from engagement-based ranking modestly reduced exposure to low-quality content but did not move users' political attitudes within the intervention window. Bagchi et al. (2024) argued the Guess et al. comparison was confounded by Meta's emergency election-period changes to the standard algorithm: "a considerable portion of the decrease reported by Guess et al. may be attributable to the temporary algorithm changes".

In the South Asian context, evidence on the role of algorithmic feeds in misinformation propagation is thinner but suggestive. Amnesty International's (2022) investigation of Meta's role in the spread of anti-Rohingya content in Myanmar concluded that Meta "substantially contributed to the serious human rights violations perpetrated against the Rohingya" (p. 7), with engagement-ranking dynamics operating in a context where "digital media literacy was extremely low" (p. 17) and local regulatory infrastructure was weak. Rahaman et al. (2026), examining Bangladesh's July 2024 uprising through 112 verified fact-checking reports, 87 news articles, and 15 interviews with journalists and fact-checkers, document a surge in politically charged disinformation amplified through social media networks that outpaced local fact-checking capacity. Their evidence does not isolate the algorithmic from the human-network contribution but supplies a strong contextual case that platform-mediated misinformation is consequential in Bangladesh.

4.3 Media literacy and individual susceptibility

Media literacy — defined in the foundational Aspen Institute report as "the ability of a citizen to access, analyze, and produce information for specific outcomes" (Aufderheide, 1993, p. 6) — has long been identified as a critical competency for democratic citizenship. In the digital context, it spans what scholars variously term news literacy, digital literacy, and critical information literacy (Mihailidis & Viotty, 2017).

The cross-individual variance in misinformation susceptibility is substantial and only partly explained by general education or intelligence. Pennycook and Rand (2019) argue that susceptibility to partisan misinformation is better explained by analytical thinking — willingness to deliberate rather than rely on intuitive judgments — than by motivated reasoning; their evidence "indicates that people fall for fake news because they fail to think; not because they think in a motivated or identity-protective way" (Pennycook & Rand, 2019). Roozenbeek et al. (2020), in five national samples (the UK, Ireland, the USA, Spain, and Mexico) during the COVID-19 pandemic, found that "higher trust in scientists and having higher numeracy skills were associated with lower susceptibility to coronavirus-related misinformation" (p. 1). The relevance for Bangladesh is direct: university-educated youth here have high general academic attainment but often minimal explicit training in source evaluation or claim verification.

Roozenbeek et al. (2022) further argue that misinformation discernment is reasonably stable across question framings and that myside bias and partisanship explain more of the variance than analytical-thinking measures do: "actively open-minded thinking ... and political ideology are both robust predictors of misinformation susceptibility, whereas classical analytic thinking ... is not" (p. 561). The picture that emerges is one of multiple, partially overlapping individual-difference predictors: analytical disposition, prior media literacy, partisan motivation, and platform-use patterns.

The concept of *epistemic vulnerability* — structural susceptibility to false belief produced by the design of one's information environment rather than by individual cognitive failure — provides a useful analytical frame (Fallis, 2021). But, as Bakshy et al. (2015) demonstrate, individual agency is not absent: users' own

choices shape exposure at least as much as algorithmic ranking does. Epistemic vulnerability is best understood as the interaction of platform architecture and individual characteristics, not as a structural determinant.

4.4 Digital news consumption in Bangladesh: the research gap

Empirical research on digital news consumption in Bangladesh is limited relative to the scale of the phenomenon. Muzykant et al. (2022) document structural deficits in media literacy that leave Bangladeshi citizens, particularly urban youth, vulnerable to platform-mediated misinformation. Haque et al. (2020) document persistent institutional constraints on Bangladesh's fact-checking ecosystem, including resource limitations and political pressure. Rahaman et al. (2026) document the surge in algorithmically amplified disinformation during the July 2024 uprising. Each of these is informative but supply-side or qualitative; none provides a quantitative measure of individual susceptibility paired with consumption-pattern data.

The algorithmic systems that mediate Bangladeshi news consumption are designed, deployed, and governed almost entirely outside the country. Empirical research on how those externally governed systems interact with the epistemic behaviour of Bangladeshi news consumers is a prerequisite for any serious national policy response. Peer-reviewed quantitative work examining the relationship between algorithmic feed exposure and measurable misinformation discernment in a Bangladeshi sample is, on a systematic JSTOR, Scopus, and Google Scholar search (January 2026), effectively absent. This study is designed to address that gap.

5. Conceptual Framework

The framework integrates platform studies, media literacy research, and the sociology of knowledge to describe the landscape of factors this study examines. It is not a linear causal pipeline. The data is cross-sectional and the relationships are best understood as a set of associations operating across three analytic levels.

Figure 1: Conceptual Framework

Algorithmic Epistemic Vulnerability		
Three interlocking analytic levels through which vulnerability is examined		
LEVEL 1 — STRUCTURAL: Platform architecture shapes the information environment		
Engagement optimisation Clicks, shares, time-on-platform	Emotional amplification Outrage and fear rewarded	Accuracy deprioritised Epistemic quality not optimised for
▼ enables		
LEVEL 2 — BEHAVIOURAL: Habitual consumption displaces critical evaluation		
Passive scrolling Mobile-first, feed-dependent	Source bypass No direct outlet engagement	Fluency bias “Familiar = credible” heuristic

▼ associated with		
LEVEL 3 — COGNITIVE (CURRENT STATE): Distribution of discernment in the study population		
Reduced verification Claim-checking habits weaker	Lower source literacy Provenance evaluation weaker	Discernment variance Distribution measured cross-sectionally
<i>Levels are mutually reinforcing, not strictly causal. Measured by: quantitative survey (Levels 1–2) and semi-structured interviews (Level 3). Moderators examined (quantitative, exploratory: media literacy training, analytical disposition, socioeconomic background; qualitative: partisanship, peer-network effects).</i>		

Source: Researcher's own framework, adapted from Pennycook & Rand (2019), Metzler & Garcia (2024), and Fallis (2021).

Level 1 — Structural. Platform architecture shapes the information environment. Engagement-ranked feeds prioritise emotionally arousing content over epistemically reliable content. This is a structural tendency well documented in the literature, not an assumption of this study.

Level 2 — Behavioural. Reliance on algorithmically curated feeds as a primary news source is associated, in existing research, with reduced practices of critical news evaluation — direct engagement with original sources, cross-referencing, and deliberate verification. Individual differences in motivation, prior beliefs, and analytical disposition moderate this association.

Level 3 — Cognitive (current state). The study examines current patterns of misinformation discernment — not degradation over time, which a cross-sectional design cannot measure, but the distribution of discernment scores, source-evaluation practices, and susceptibility to engagement-cued content as they exist in the study population at the time of measurement.

The framework treats motivated reasoning, partisanship, prior media literacy training, peer network effects, and socioeconomic factors as substantive moderators that may explain variance in discernment independently of platform reliance. The quantitative phase tests the relative predictive contribution of the variables it measures directly (prior media literacy training, analytical disposition via the adapted AOT scale, and socioeconomic background); partisanship, motivated reasoning, and peer-network effects are explored qualitatively in Phase 2 rather than measured quantitatively. The framework is the researcher's own integrative reading of Pennycook & Rand (2019), Metzler & Garcia (2024), and Fallis (2021); it is offered as a way of organising the analysis, not as a tested model.

6. Methodology

6.1 Research design

This study uses a sequential explanatory mixed-methods design (Creswell & Plano Clark, 2025). The quantitative phase establishes the patterns and correlates of misinformation discernment across the sample. The qualitative phase explores how participants make sense of those patterns in their own

terms. The design is cross-sectional; no causal inferences are drawn.

The study will be pre-registered on the Open Science Framework after the N = 60 pilot has frozen the adapted instruments and before any main-phase data collection begins; the final Bengali instruments will be attached to the registration as components. The pre-registration will specify the hypotheses (H1–H4, section 3), the primary outcome (MIST-20 overall discernment score), the robustness measures (adapted AOT scale and Bangladesh headline task), the convergence-handling rule (section 6.3), the full regression models with all covariates, the participant-exclusion and missing-data rules, the multiple-comparison policy, the explicit designation of confirmatory versus exploratory analyses, the headline-task exclusion rule, and the contingency plan for qualitative sampling under both confirmed and disconfirmed quantitative findings (section 6.5).

6.2 Study setting and target population

The study uses a purposive multi-site design across six universities in Dhaka, three private and three public. The selection logic is theoretically motivated rather than convenience-based: digital-capital variables (home internet quality, primary device, schooling medium) and broader socioeconomic variables vary systematically across institution type. Including both maximises variation on the moderator variables this study examines.

The six sites and the selection rationale:

Private universities:

1. **University of Liberal Arts Bangladesh (ULAB)** — established mid-tier private, English-medium, central Dhaka. The lead researcher has direct institutional access.
2. **BRAC University** — higher-tier private with strong research infrastructure; broad disciplinary mix.
3. **North South University (NSU)** — large-enrolment private with a mixed socioeconomic intake spanning the upper-middle to lower-middle bracket of the private sector.

Public universities:

4. **University of Dhaka (DU)** — elite national public university with the largest and most diverse undergraduate population.
5. **Jahangirnagar University (JU)** — residential public university with a distinct campus culture and mixed socioeconomic intake.
6. **Bangladesh University of Engineering and Technology (BUET)** — elite technical public with predominantly Bangla-medium intake and a different disciplinary profile from DU and JU.

Institution type is entered as a binary predictor (private/public) in the regression. Site is entered as a fixed effect to absorb between-site variance that institution type does not explain. The institution-type

effect is interpretable only after controlling for individual-level socioeconomic variables (see 6.3); the proposal does not claim that any institution-type effect reflects institutional culture rather than student composition until that test has been run.

The target population at each site is undergraduate students aged 18–28 who use Facebook, YouTube, or AI-mediated news sources at least once per week. No restriction is placed on faculty, year of study, or gender.

6.3 Phase 1: Quantitative survey

Sample, recruitment, and power

Target sample: 300–360 students across the six sites, with 50–60 per site. Recruitment is conducted by the lead researcher — a student peer with no teaching, grading, or supervisory authority over participants — using standardised recruitment scripts; if project funding is secured, recruitment is delegated to a research assistant. Stage one stratifies by institution type (private/public) and then by site. Stage two stratifies within each site by faculty to ensure disciplinary variation. The sampling frame at each site is the registrar's enrolment roster, accessed under a data-sharing agreement negotiated with each institution before data collection. Where institutional access is denied at a site, that site is replaced and the limitation noted explicitly.

A single compensation policy — a 500 BDT mobile recharge or bKash transfer if funding is secured, or uncompensated voluntary participation if not — is fixed once before main-phase data collection and applied uniformly to all participants; the chosen policy is stated identically in the recruitment materials and the consent form.

Power: at $f^2 = 0.05$ (a more empirically defensible "small" effect in media-effects regressions) with 12 predictors at $\alpha = .05$ and power = .80, the required N is approximately 300, matching the recruited target. The $f^2 = 0.05$ figure refers to the omnibus model; the focal single-coefficient test of H1 requires approximately $N = 170$ at the same parameters, so the recruited target provides headroom. For a single interaction term, $N = 300$ provides 80% power only for incremental $f^2 \geq .027$, which is why moderation analyses are designated exploratory (section 3). The conventional medium-effect assumption ($f^2 = 0.15$) yields a minimum N of 118; the higher target provides headroom for missing data, attrition, and subgroup analyses by institution type. Site-level inference is acknowledged as limited: a between-site contrast at $N \approx 50$ vs $N \approx 50$ detects Cohen's $d \geq .56$, which is moderate-to-large; smaller site-level effects cannot be reliably detected and are not claimed.

Survey instrument

The survey has four sections.

Section A — demographics and socioeconomic background. Age, gender, faculty, year of study, prior

media literacy education, and four socioeconomic / digital-capital variables: (i) home internet access quality on a 4-point scale, (ii) primary device used for news consumption, (iii) schooling medium (Bangla / English / mixed), and (iv) parental education level (highest of two parents), used as a proxy for household socioeconomic status. These are entered as controls in the regression to test whether observed institution-type effects survive socioeconomic adjustment.

Section B — news consumption patterns. Frequency-of-use and proportion-of-news items are collected separately for three modalities:

- *Facebook algorithmic feed.* Defined as news encountered through scrolling the main Facebook feed, including posts from pages, friends, and groups, with no direct visit to a news outlet.
- *YouTube feed.* Defined as news encountered through YouTube’s algorithmically recommended surfaces — the home feed, Up Next recommendations, and Shorts — including news clips, talk shows, and news-style channels, without deliberate search for a specific outlet. Modality selection follows a news-gateway criterion, not a user-count criterion: Facebook and YouTube are the two platforms with documented mass reach in Bangladesh (Kemp, 2025) and video platforms are a documented and growing news source for under-35s (Newman et al., 2025), while no Bangladesh-specific evidence documents TikTok as a news gateway and its content ecosystem is predominantly entertainment-oriented. TikTok news exposure is therefore captured by a single exploratory frequency item rather than as a confirmatory modality.
- *AI-mediated news exposure.* Operationally defined as news content obtained from (a) explicit queries to ChatGPT, Gemini, Claude, or comparable assistants; (b) AI Overview blocks at the top of Google search results; (c) lock-screen or in-app AI summaries (Pixel, iOS, news apps). Each sub-component is captured separately and a composite index constructed only after exploratory analysis confirms the sub-components covary.

Additional items measure direct news consumption (visiting a news outlet's site or app), self-reported awareness of algorithmic curation, and political-news interest.

Section C — misinformation discernment (the primary outcome and two robustness checks).

The primary outcome is the **MIST-20** (Maertens et al., 2024), the 20-item form of the first psychometrically validated misinformation susceptibility test, adapted into Bengali using the four-stage procedure described below. The MIST-20 yields three scores: real-news detection, fake-news detection, and overall veracity discernment; the overall discernment score is the primary outcome.

Two robustness measures accompany the MIST-20:

- An **adapted shortened AOT scale** (seven items, drawn from the Haran et al. (2013) shortened version of Stanovich and West's Actively Open-Minded Thinking scale), measuring disposition toward evidence-based reasoning.

- A **Bangladesh headline discernment task** of 20 items (10 accurate, 10 fabricated), constructed for ecological validity in the local context. Fabricated items are drawn from genuine misinformation events documented in Bangladeshi fact-checking records (Rumor Scanner, Fact Watch), excluding the Azimpur incident and any incident the lead researcher has previously reported on or fact-checked, including in his work with FactWatch (full exclusion list documented in the pre-registration). Each headline is accompanied by a 5-point share-intent item ("How likely would you be to share this with a friend or family member?"), which provides a partial behavioural supplement to the stated-judgment task.

Convergence-handling rule (pre-registered): the MIST-20 overall discernment score is the primary outcome of all hypothesis tests. The adapted AOT scale and the Bangladesh headline task are reported as robustness checks. If the three measures converge (pairwise $r \geq .40$), a composite z-score average is reported as a supplementary outcome. If they diverge (pairwise $r < .40$), each is analysed separately and the divergence is discussed substantively, not treated as a measurement failure.

Section D — media literacy and AI literacy. Self-reported prior media literacy training; familiarity with how AI language models generate text; understanding of their limitations.

Scale adaptation procedure

The MIST-20, the AOT short scale, and the Bangladesh headline task are adapted using a four-stage procedure. First, items are reviewed by the researcher and supervisor for conceptual equivalence; items with no equivalent local referent are replaced with contextually equivalent items, with the substitution rules documented. Second, instruments are forward-translated into Bengali by a volunteer independent bilingual translator (not the lead researcher; e.g., a bilingual faculty member or graduate student, acknowledged in the final paper), back-translated into English by a second volunteer independent bilingual translator, with discrepancies reconciled by both translators in consultation with the supervisor. Third, adapted instruments are pilot-tested with 60 students (not 20, to support stable item-level decisions); items where accuracy clusters above 90 percent correct or below 30 percent correct are flagged for replacement. Fourth, internal consistency (Cronbach's α) is calculated on the pilot data; items reducing α

materially are revised. The full adaptation log is reported in the methods section of the final paper and posted with the pre-registration.

Analysis

Descriptive statistics characterise the sample's consumption patterns and discernment scores, reported overall and disaggregated by institution type. The confirmatory models are specified as follows. Model 1 (H1) regresses MIST-20 overall discernment on the composite feed-reliance index plus eleven covariates — age, gender, faculty, year of study, prior media literacy training, political-news interest, home internet quality, primary device, schooling medium, parental education, and institution type — with site entered

as fixed effects. Model 2 (H2) replaces the composite index with the three modality scores (Facebook feed, YouTube feed, AI-mediated exposure); the Facebook and YouTube coefficients are Holm-corrected as one confirmatory family, and the AI-mediated exposure coefficient is reported as exploratory with no directional prediction. Model 3 (H3) re-estimates Model 1 without the four socioeconomic / digital-capital variables; the institution-type coefficient is reported with and without adjustment to make the contribution of student composition versus institutional context visible. Effect sizes (Cohen's f^2) accompany all coefficients. Multiple-comparison policy: H1 is a single test at $\alpha = .05$; the two confirmatory H2 tests (Facebook, YouTube) form one Holm-corrected family; the AI-mediated exposure coefficient is exploratory; analyses of the two robustness outcomes are reported as robustness information, not as additional confirmatory tests. Exploratory analyses — the moderation interactions, share-intent discernment (mean share-intent for accurate minus fabricated headlines, correlated with MIST-20 scores), algorithmic-awareness contrasts, the AI-exposure sub-component composite, the exploratory TikTok news-frequency item, and any post-hoc analysis of institutional characteristics — are labelled exploratory in all reporting. All inferences are reported as correlational; no causal claims are drawn from the regression.

Exclusion and missing-data rules

Participant-level exclusions, applied before analysis with counts reported: survey completion time below 40 percent of the pilot median; failure of either of two embedded attention-check items; zero response variance across the MIST-20 block; duplicate participation detected via institutional e-mail or phone number; or a missing primary outcome. Item-level: the pre-registered headline-task exclusion list (the Azimpur incident and any incident the lead researcher has previously reported on or fact-checked, including in his work with FactWatch).

Missing data: confirmatory models use complete cases if missingness across model variables is below 5 percent; otherwise covariates are multiply imputed by chained equations ($m = 20$), with complete-case results reported as a sensitivity analysis. The primary outcome is never imputed.

6.4 Phase 2: Qualitative interviews

Sample and recruitment

After the quantitative phase, 18 participants are recruited for semi-structured interviews using purposive sampling, three per site. Within each site, participants are selected for variation across MIST-20 discernment level (high, medium, low) and platform reliance pattern. Outlier participants — those with high platform reliance but high discernment — are specifically prioritised across sites to surface potential protective factors.

A pre-registered contingency: if the quantitative phase does not yield a significant association between platform reliance and discernment, qualitative recruitment shifts to maximum-variation sampling across discernment levels, with the interview guide adjusted to probe how high-discernment students reason about platform content regardless of consumption volume.

Interview guide

Interviews are semi-structured, approximately 45–60 minutes, default English with Bengali permitted for sensitive material (ULAB and most private sites operate primarily in English; DU/JU/BUET participants may prefer Bengali). The guide covers five thematic areas: news consumption habits and platform experience; awareness of algorithmic curation; credibility-evaluation practices, including walkthroughs of recent examples; misinformation experience, including instances of successful identification; and perceived barriers and enablers of critical news evaluation.

Transcription, translation, and coding

Audio is recorded with informed consent. Transcription is done verbatim by the lead researcher in the language of the interview, with all recordings retained for audit. Bengali transcripts are forward-translated by a volunteer independent bilingual translator; the lead researcher reconciles translation

choices in consultation with the translator and a bilingual second coder. Critically, the bilingual second coder works from the Bengali source transcript, not from the English translation, to detect any interpretive drift introduced by translation. A 20 percent double-coded subset checks coding reliability (target Cohen's $\kappa \geq .70$).

Thematic analysis follows Braun and Clarke's (2006) six-phase framework. Analysis is guided by but not limited to the conceptual framework, allowing for emergent themes. Original Bengali excerpts are retained in the data record alongside translations. Member checks on translated quotes are offered to all participants who use Bengali in the interview.

6.5 Integration of quantitative and qualitative data

Quantitative and qualitative findings are integrated at the interpretation stage using a connecting strategy: the quantitative results identify what patterns exist; the qualitative findings explore how participants make sense of those patterns. Two pre-specified integration scenarios:

Scenario A (predicted association confirmed): qualitative sampling targets outliers (high reliance, high discernment) and interviews probe what protective factors these participants describe.

Scenario B (predicted association weak or absent): qualitative sampling shifts to maximum variation on discernment, and interviews probe the reasoning practices of high-discernment students irrespective of consumption volume.

Where quantitative and qualitative findings diverge in either scenario, the divergence is treated as analytically significant, not as a failure of one method.

6.6 Reflexivity and structural mitigations

The lead researcher's prior professional experience as a journalist who reported on the Azimpur misinformation incident, and his current fact-checking work with FactWatch, create an analytical asset and a potential bias. The FactWatch affiliation also creates a source-selection risk for the headline task, which draws fabricated items from published fact-checking archives. The asset is contextual knowledge and network access. The risk is that the research design and analysis are unconsciously shaped toward confirming prior observations.

The reflexivity is addressed through writing — a positionality statement, a reflective journal — and through structural separation of roles where feasible within a largely solo, unfunded project, supported by supervisor oversight and named volunteer collaborators:

1. Recruitment uses standardised scripts; the lead researcher is a student peer with no grading or supervisory authority over participants, and recruitment is delegated to a research assistant if funding is secured.
2. Survey data collection is administered primarily through an online platform accessed independently via link or QR code; where online completion is impractical, an identical paper version is used, with survey mode recorded for every respondent, paper responses double-entered, and a pre-specified sensitivity analysis (Model 1 re-estimated on online-only responses) if paper responses exceed 10 percent of the sample.
3. Interview transcription is done verbatim by the lead researcher, with all recordings retained so that transcripts can be audited against audio; a volunteer bilingual checker audits a sample.
4. Forward translation of Bengali transcripts is done by a volunteer independent bilingual translator (not the lead researcher).
5. The volunteer bilingual second coder of qualitative data works from the Bengali source, not from the English translation.
6. A subset of quantitative analyses (the primary regression model) is replicated by the supervisor independently before findings are reported.
7. The Azimpur incident and any incident the lead researcher has previously reported on or fact-checked (including at FactWatch) are excluded from the headline task by pre-registered rule.

These separations make the researcher's interpretive choices visible at every stage of the pipeline and reduce the surface area for unconscious confirmation bias. They do not eliminate the risk, and the final paper carries an explicit reflexivity statement disclosing the researcher's professional background and these mitigations.

6.7 Ethical considerations

All participants receive a plain-language information sheet explaining the study's purpose, procedures, and their right to withdraw at any time without consequence. Written informed consent is obtained

before survey completion and verbal recorded consent before interviews. Survey data is stored anonymously; interview participants are assigned pseudonyms. Data is stored on encrypted institutional storage and accessed only by the research team. Ethical approval is sought from ULAB's research ethics committee. For the five partner institutions, the ULAB approval documentation is shared with relevant departmental contacts; where a partner institution requires separate clearance, that process is initiated in parallel during Weeks 2–4 of the workplan.

6.8 Limitations

This study has seven limitations that should be acknowledged explicitly.

Cross-sectional design. The design permits correlation analysis only; it cannot establish causal direction or change over time.

Sample bounds. The study is bounded to urban Dhaka university students. Findings should not be generalised to rural Bangladesh, non-university populations, or university students outside Dhaka without further research.

Multi-site coordination risk. Execution is planned as a largely solo project, which increases coordination burden; the standardised protocols, volunteer roles, and supervisor oversight specified in section 6.6 mitigate but do not eliminate this. Six-site execution introduces delays in institutional access, variation in site coordinator engagement, and the possibility that access is denied at one or more sites. The design includes a one-week preparation buffer (see workplan) and a site-replacement contingency; the risk cannot be eliminated.

Self-report and stated-judgment. Self-reported news consumption is subject to social desirability and recall limitations. The headline task is a stated-judgment measure; share-intent items provide a partial behavioural supplement but ecological validity remains lower than for actual platform behaviour.

New / adapted instruments. The Bangladesh headline task is newly constructed; the MIST-20 and AOT short scale are adapted into Bengali. The N = 60 pilot supports basic item functioning checks but is not sufficient for full psychometric validation. The reliability of these instruments in Bangladeshi samples is reported transparently and treated as a contribution to the methodological literature rather than as a settled measurement.

Researcher prior involvement. The lead researcher previously reported on, and as a FactWatch contributor has fact-checked, misinformation incidents relevant to the study. The structural mitigations in 6.6 are designed to manage but not eliminate this risk.

Site-level inference is limited. $N \approx 50$ per site supports descriptive characterisation and contrasts of moderate-to-large effects ($d \geq .56$); finer-grained between-site comparisons are not powered.

7. Workplan

The 22-week timeline runs from July to early December 2026. The two-week buffer relative to the prior plan absorbs preparation-phase coordination risk (institutional access, site coordinator confirmation, instrument piloting). The workplan has four phases: Preparation (Weeks 1–6), Data Collection (Weeks 7–14), Analysis (Weeks 13–18), and Write-Up (Weeks 18–22), with intentional overlap between adjacent phases.

[illegible][illegible]

16	Write-up and discussion								✓	✓	✓
17	Supervisor review and revision									✓	✓
18	Final draft and submission										✓

8. Expected Contributions

The study aims to make four contributions, with scope appropriate to a single-city, cross-sectional, mixed-methods design.

First, it produces the first comparative quantitative dataset on algorithmic news consumption and misinformation discernment across private and public university students in Dhaka — a sample frame underrepresented in the South Asian digital media literature and one that prior single-institution studies cannot address.

Second, by disaggregating consumption across Facebook feed, YouTube feed, and AI-mediated news exposure, and by controlling for four socioeconomic and digital-capital variables, the study produces more precise estimates of which specific consumption patterns are associated with discernment than prior single-platform studies have achieved.

Third, the institution-type contrast, run with and without socioeconomic adjustment, directly tests whether observed institutional differences in discernment reflect student composition or something specific to the institutional context — a question with direct implications for where media-literacy interventions should be targeted.

Fourth, the study contributes a set of locally adapted instruments (Bengali MIST-20, adapted shortened AOT, Bangladesh headline task) and a methodological template (independent translation, role-separated reflexivity, MIST-20 as primary outcome with two robustness checks) that other Bangladeshi institutions can adapt or contest.

The study is positioned as foundational empirical work for a future research agenda this study cannot itself pursue: longitudinal tracking of consumption–discernment associations, nationally representative sampling across Bangladesh's tertiary system, and experimental manipulation of feed exposure. The present study establishes the baseline comparative dataset and the locally validated instruments that such future work requires.

AI Use Statement

During the drafting of this proposal, Large Language Models (LLMs) were used strictly for language polishing, structural feedback, and copy-editing, including verification of citations against source documents. All core conceptualisation, research design, literature synthesis, and

analytical planning are the sole work of the human researcher, who takes full responsibility for the accuracy of all content and citations.

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